Eigenvalues and inverse problems

Johnson's derivative-realizability conjecture

Difficulty: challenging **Importance:** interesting to the community

Status: open.

Problem statement

For every integer $n \geq 5$ and every real entrywise-nonnegative matrix $A \in \mathbb{R}^{n\times n}$, define $p_A(z) = \det(zI - A)$. Must there exist an entrywise-nonnegative $B \in \mathbb{R}^{(n-1)\times(n-1)}$ such that

$$\det(zI - B) = \frac{1}{n} p_A'(z) \qquad \text{as polynomials in } z?$$

Equivalently, the critical points of the characteristic polynomial, counted with multiplicity, would themselves form a realizable spectrum of the smaller order. Neither A nor B is assumed symmetric or diagonalizable. The realization must have exactly order $n-1$; allowing arbitrary additional zero eigenvalues changes the problem. This would provide a dimension-reduction operation for nonnegative spectral realization.

References

Hoover, McCormick, Paparella, and Thrall, *On the realizability of the critical points of a realizable list*, Conjecture 1.2, p. 2, and §6. The paper credits the conjecture to Johnson and records the Cronin–Laffey low-order results.

Status check

Searches for *Johnson conjecture derivative nonnegative matrix characteristic polynomial proof counterexample*, *1712.05454 2026*, and *Monov conjecture solved* found no general resolution. The source proves several classes and records the solved cases $n \leq 4$, and $n \leq 6$ with $\operatorname{tr} A = 0$. Nonnegative power sums alone are a different hypothesis. Monov's weaker moment conjecture is not separately counted here.